Difference-in-Differences

Methodology Reading List

A thematically organized bibliography covering the foundational literature, core identification theory, and frontier methodological developments in the DiD/TWFE tradition. Papers marked † are particularly recommended as entry points for each section.

I. Foundational Works

Ashenfelter, O., & Card, D. (1985). Using the longitudinal structure of earnings to estimate the effect of training programs. *Review of Economics and Statistics*, *67*(4), 648–660.

> One of the earliest formal DiD applications; introduces the "Ashenfelter's Dip" phenomenon and motivates the parallel trends assumption.

† Card, D., & Krueger, A. B. (1994). Minimum wages and employment: A case study of the fast-food industry in New Jersey and Pennsylvania. *American Economic Review*, *84*(4), 772–793.

> The canonical 2×2 DiD application; seminal in popularizing quasi-experimental design and in framing the empirical debate over identification assumptions.

† Bertrand, M., Duflo, E., & Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, *119*(1), 249–275.

> Identifies the serial correlation problem in long panel DiD; demonstrates severe under-rejection of the null under naive standard errors. Recommends cluster-robust inference. Essential pre-reading for any applied work.

Duflo, E. (2001). Schooling and labor market consequences of school construction in Indonesia. *American Economic Review*, *91*(4), 795–813.

> Classic application illustrating Intent-to-Treat estimation using natural experiments within a DiD framework.

II. Identification and the Potential Outcomes Framework

† Imbens, G. W., & Angrist, J. D. (1994). Identification and estimation of local average treatment effects. *Econometrica*, *62*(2), 467–475.

> Establishes the LATE framework; the theoretical foundation for Fuzzy DiD and IV-based compliance corrections.

Heckman, J. J., Ichimura, H., & Todd, P. (1997). Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *Review of Economic Studies*, *64*(4), 605–654.

> Connects DiD to matching estimators; clarifies how conditioning on observables relaxes identification assumptions and motivates the conditional PTA literature.

Abadie, A. (2005). Semiparametric difference-in-differences estimators. *Review of Economic Studies*, *72*(1), 1–19.

> Derives the IPW-DiD estimator under conditional PTA; foundational for the doubly robust and machine-learning extensions that follow.

III. Parallel Trends: Testing, Validation, and Sensitivity

† Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, *4*(3), 115–132.

> Demonstrates that conditioning final inference on non-significant pre-trends distorts the sampling distribution of post-treatment estimates, producing size distortions and reduced power. The key epistemological caution for event-study practice.

† Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, *90*(5), 2555–2591.

> Develops "Honest DiD": a sensitivity analysis framework permitting bounded, researcher-calibrated deviations from strict parallel trends based on observed pre-trend magnitudes.

Marcus, M., & Sant'Anna, P. H. (2021). The role of parallel trends in event study settings. *Journal of the Association of Environmental and Resource Economists*, *8*(2), 235–275.

> Analyzes how the choice of comparison group (never-treated vs. not-yet-treated) governs the strength of the PTA and the identification of causal parameters in staggered settings.

Freyaldenhoven, S., Hansen, C., & Shapiro, J. M. (2019). Pre-event trends in the panel event-study design. *American Economic Review*, *109*(9), 3307–3338.

> Proposes an IV-based correction for settings where observed pre-trends reflect genuine confounding; separates spurious anticipation from true parallel trends violations.

IV. TWFE Decomposition and the Staggered Adoption Problem

† Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, *225*(2), 254–277.

> Proves that the TWFE coefficient is a variance-weighted average of all possible 2×2 sub-comparisons, including "forbidden" comparisons that use already-treated units as controls.

† de Chaisemartin, C., & d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, *110*(9), 2964–2996.

> Characterizes the negative weighting problem under TWFE and proposes heterogeneity-robust estimators for settings with treatment switching.

Baker, A. C., Larcker, D. F., & Wang, C. C. Y. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, *144*(2), 370–395.

> Demonstrates the severity of TWFE bias in corporate finance empirical settings; benchmarks five modern estimators against each other and against naïve TWFE across hundreds of published studies.

Jakiela, P. (2021). Simple diagnostics for two-way fixed effects. *arXiv* preprint 2103.13229.

> Derives simple graphical and numerical diagnostics to detect when TWFE negative weights are large enough to materially bias estimates; useful first-pass tool before choosing a staggered estimator.

V. Modern Estimators for Staggered Adoption

† Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230.

> Proposes estimating cohort-time ATT(g,t) using clean comparison groups (never- or not-yet-treated), with flexible aggregation. The most widely adopted modern staggered DiD estimator.

Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, *225*(2), 175–199.

> Derives the interaction-weighted (IW) estimator as a heterogeneity-robust alternative to TWFE for dynamic event studies under staggered timing.

† Borusyak, K., Jaravel, X., & Spiess, J. (2024). Revisiting event study designs: Robust and efficient estimation. *Review of Economic Studies*, *91*(3), 1324–1361.

> Imputation approach: fits TWFE exclusively on untreated observations, then imputes counterfactuals for treated units. Achieves robustness to treatment effect heterogeneity with efficiency gains over cohort-based methods.

† Roth, J., & Sant'Anna, P. H. C. (2023). Efficient estimation for staggered rollout designs. *Journal of Political Economy Microeconomics*, *1*(4), 669–709.

> Derives the semiparametrically efficient estimator nesting CS2021, BJS, and related approaches, under the assumption that treatment timing is quasi-randomly assigned. Provides both t-based and permutation inference. Efficiency gains over existing estimators can be substantial.

Cengiz, D., Dube, A., Lindner, A., & Zipperer, B. (2019). The effect of minimum wages on low-wage jobs. *Quarterly Journal of Economics*, *134*(3), 1405–1454.

> Prominent application of the stacked DiD design, restructuring data into cohort-specific clean sub-samples to eliminate invalid comparison groups by construction.

† Roth, J., Sant'Anna, P. H., Bilinski, A., & Poe, J. (2024). What's trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics*, *235*(2), 105432.

> Comprehensive synthesis of the "New DiD" literature; the recommended starting point for practitioners seeking unified guidance on staggered adoption, pre-trend testing, and modern estimator selection.

VI. Conditional Parallel Trends and High-Dimensional Covariates

Sant'Anna, P. H., & Zhao, J. (2020). Doubly robust difference-in-differences estimators. *Journal of Econometrics*, *219*(1), 101–122.

> Combines outcome regression and propensity score weighting; consistent if either component is correctly specified. The practical benchmark for covariate-adjusted DiD.

† Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrics Journal*, *21*(1), C1–C68.

> Introduces DML: machine learning estimation of nuisance parameters with cross-fitting and Neyman-orthogonalization to eliminate regularization bias. The theoretical foundation for ML-based DiD.

Chang, N. C. (2020). Double/debiased machine learning for difference-in-differences models. *Econometrics Journal*, *23*(2), 177–191.

> Adapts DML to the DiD setup; non-parametrically accommodates high-dimensional, time-varying confounders while supporting valid uniform inference on the ATT.

VII. Fuzzy DiD and Imperfect Compliance

de Chaisemartin, C., & d'Haultfœuille, X. (2015). Fuzzy differences-in-differences. *Review of Economic Studies*, *85*(2), 999–1028.

> Formalizes Fuzzy DiD by using the original policy assignment as an instrument; derives the Wald-DiD estimator to recover the LATE under imperfect compliance and control group contamination.

VIII. Hybrid Methods

† Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies. *Journal of the American Statistical Association*, *105*(490), 493–505.

> Introduces the synthetic control method; the methodological precursor to SDID and the standard approach for single treated-unit settings.

† Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, *111*(12), 4088–4118.

> Unifies SCM and DiD via unit and time weights with regularized fixed effects; robust to unobserved heterogeneous trends that violate the strict PTA. Formally demonstrates efficiency dominance over both SCM and DiD in relevant settings.

Xu, Y. (2017). Generalized synthetic control method: Causal inference with interactive fixed effects models. *Political Analysis*, *25*(1), 57–76.

> Extends the synthetic control framework by estimating an interactive fixed effects model to construct counterfactuals; bridges factor models, SCM, and DiD into a unified latent factor approach.

† Grembi, V., Nannicini, T., & Troiano, U. (2016). Do fiscal rules matter? *American Economic Journal: Applied Economics*, *8*(3), 1–30.

> Introduces the canonical difference-in-discontinuities (DiDC) design: a policy changes the RD threshold across time, allowing one to difference out both time-invariant RD-based selection and aggregate time trends simultaneously.

Hausman, C., & Rapson, D. S. (2018). Regression discontinuity in time: Considerations for empirical applications. *Annual Review of Resource Economics*, *10*, 533–552.

> Examines "RD in time" (RDIT) designs where the treatment assignment rule changes at a known calendar date; discusses when RDIT and DiD assumptions coincide and where they diverge, and characterizes the relevant validity threats.

Lalive, R. (2008). How do extended benefits affect unemployment duration? A regression discontinuity approach. *Journal of Econometrics*, *142*(2), 785–806.

> Influential application combining RD variation in benefit duration rules with temporal policy changes; illustrates how geographic and temporal discontinuities can be jointly exploited within a DiD-style framework.

IX. Distributional, Non-linear, and Continuous Treatment Extensions

† Athey, S., & Imbens, G. W. (2006). Identification and inference in nonlinear difference-in-differences models. *Econometrica*, *74*(2), 431–497.

> Develops the Changes-in-Changes (CiC) model under a non-separable structure, permitting recovery of the full counterfactual distribution without additive separability or scale restrictions.

Callaway, B., & Li, F. (2019). Quantile treatment effects in difference-in-differences models with panel data. *Quantitative Economics*, *10*(4), 1579–1618.

> Extends DiD to the quantile estimand (QTET) under copula stability; evaluates heterogeneous distributional impacts across the treated outcome distribution.

Wooldridge, J. M. (2023). Simple approaches to nonlinear difference-in-differences with panel data. *Econometrics Journal*, *26*(3), C31–C66.

> Proposes pooled QMLE strategies for non-negative, binary, and fractional outcomes under a generalized index formulation of parallel trends, circumventing the incidental parameters problem.

† Callaway, B., Goodman-Bacon, A., & Sant'Anna, P. H. (2024). Difference-in-differences with a continuous treatment. *Journal of Econometrics*, *239*(2), 105620.

Identifies severe selection bias and interpretational ambiguities in linear continuous TWFE; recommends non-parametric dose-response estimation under Generalized Parallel Trends.

X. Comparative Studies and Simulation Evidence

Papers that systematically evaluate, benchmark, or compare multiple estimators across simulation designs or large-scale replication studies.

† Baker, A. C., Larcker, D. F., & Wang, C. C. Y. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, *144*(2), 370–395.

> Benchmarks five modern staggered DiD estimators (CS, SA, BJS, stacked, CSDID) against naïve TWFE across hundreds of published corporate finance studies via simulation and re-estimation; quantifies practical bias magnitudes.

Roth, J., Sant'Anna, P. H., Bilinski, A., & Poe, J. (2024). What's trending in difference-in-differences? *Journal of Econometrics*, *235*(2), 105432.

> Provides a unified comparative framework for existing staggered estimators, pre-trend tests, and sensitivity methods; includes formal equivalence results and practical decision guidance.

† Liu, L., Wang, Y., & Xu, Y. (2024). A practical guide to counterfactual estimators for causal inference with time-series cross-sectional data. *American Journal of Political Science*, *68*(1), 160–176.

> Systematically compares DiD, SCM, generalized synthetic control, and matrix completion methods on simulated and real data; provides a decision tree for method selection based on data structure and identifying assumptions.

Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230.

> The simulation appendix provides direct numerical comparisons of CS2021, TWFE, and Sun–Abraham across designs with heterogeneous treatment effects; illustrates conditions under which TWFE bias is severe.

Wing, C., Simon, K., & Bello-Gomez, R. A. (2018). Designing difference in difference studies: Best practices for public health policy research. *Annual Review of Public Health*, *39*, 453–469.

> Reviews the principal design choices in applied DiD—comparison group selection, pre-period length, functional form, and inference—from a policy evaluation standpoint; useful practical companion to the theoretical literature.

Conley, T. G., & Taber, C. R. (2011). Inference with "difference in differences" with a small number of policy changes. *Review of Economics and Statistics*, *93*(1), 113–125.

Addresses the inference problem when only a small number of groups receive treatment; proposes an asymptotic framework treating the number of treated groups as fixed and shows that conventional cluster-robust inference can be severely misleading in such settings.

† Angrist, J. D., & Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

Chapter 5 provides the most widely cited textbook treatment of DiD and TWFE; essential background for understanding what the "New DiD" literature diagnoses and corrects.

† Roth, J., Sant'Anna, P. H., Bilinski, A., & Poe, J. (2024). *(See Section V and XI above.)*

Callaway, B. (2023). Difference-in-differences for policy evaluation. In *Handbook of Labor, Human Resources and Population Economics.* Springer.

Accessible survey of modern DiD methods with an emphasis on practical implementation; useful bridge between theory and applied research.

Last updated: May 2026